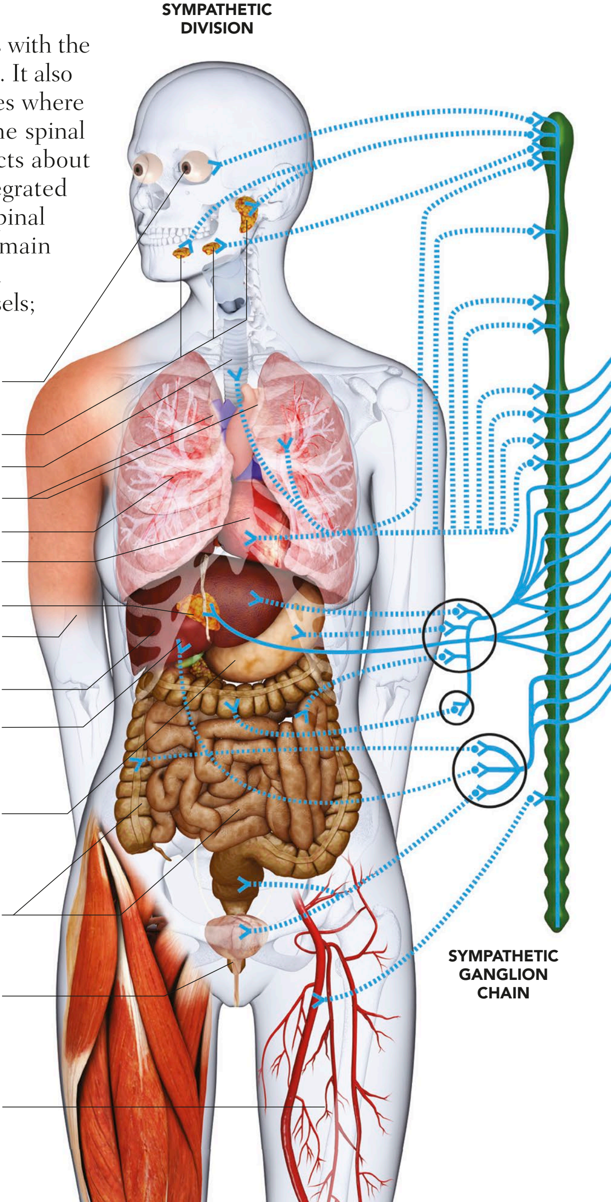
AUTONOMIC NERVOUS SYSTEM

THE AUTONOMIC NERVOUS SYSTEM (ANS) MAINTAINS CONSTANT CONDITIONS WITHIN THE BODY, A PROCESS KNOWN AS HOMEOSTASIS. MOST OF ITS ACTIVITY IS INDEPENDENT (AUTONOMIC) OF THE CONSCIOUS MIND.

AUTOMATIC FUNCTIONS

The ANS shares some nerve structures with the central and peripheral nervous systems. It also has chains of ganglia (clusters of nerves where axons communicate) either side of the spinal cord. The sensory information it collects about organs and internal activities are integrated by the hypothalamus, brainstem, or spinal cord. It sends motor signals to three main destinations: the involuntary smooth muscles of many organs and blood vessels; cardiac muscle; and certain glands.



TWO DIVISIONS

There are two divisions in the ANS: the sympathetic and parasympathetic. The ganglia of the sympathetic division are arranged into two ganglion chains, one either side of the spinal column (only one is shown here). The ganglia of the parasympathetic ANS are inside organs (see diagram). Only skin and blood vessels receive nerve messages from all positions on the cord.